

INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR (IIT Gandhinagar)



Assignment Report

M.Tech in Integrated Circuit Design and Technology (ICDT)

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Subject Name:	Analog IC Design Lab
Subject Code:	EE 653
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TASK 1: Designing Folded Cascode OTA with NMOS Load

Design a Folded Cascode Operational Amplifier with a single-ended output for the following specifications. Use the Current reference to bias the transistors wherever required. Run the simulations for all corners (tt, ss, ff, sf, fs) and report the specs obtained for all corners in tabular format.

- Differential Gain > 80 dB
- CMRR > 100 dB
- ICMR: $0.8 - 1.8$ V
- Max. Output Swing: 1 V
- Power $< 20 \mu\text{W}$
- GBW > 10 MHz

Schematic of Folded Cascode with NMOS load

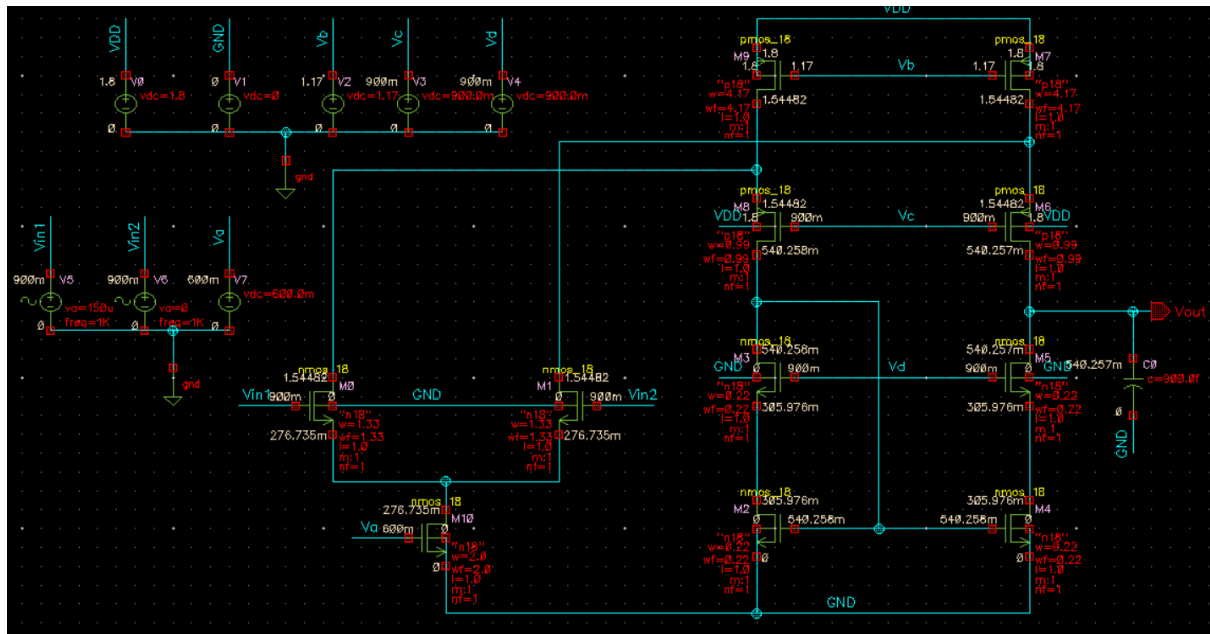


Figure 1: component parameters

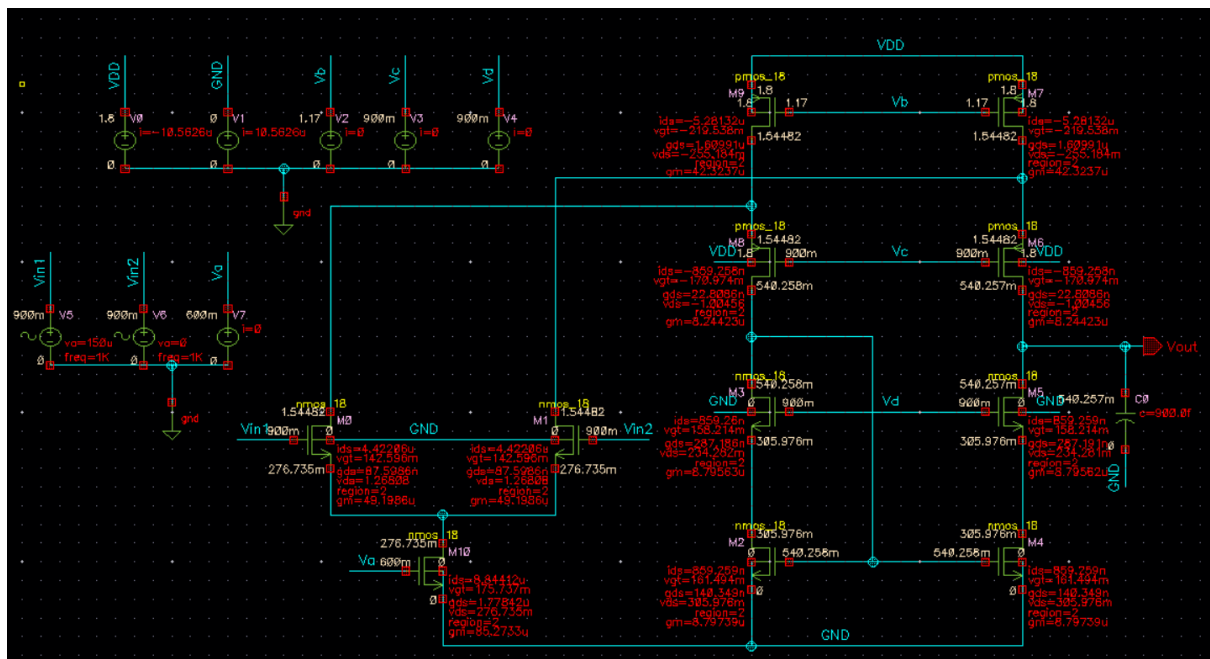


Figure 2: DC Parameters

Simulation Results

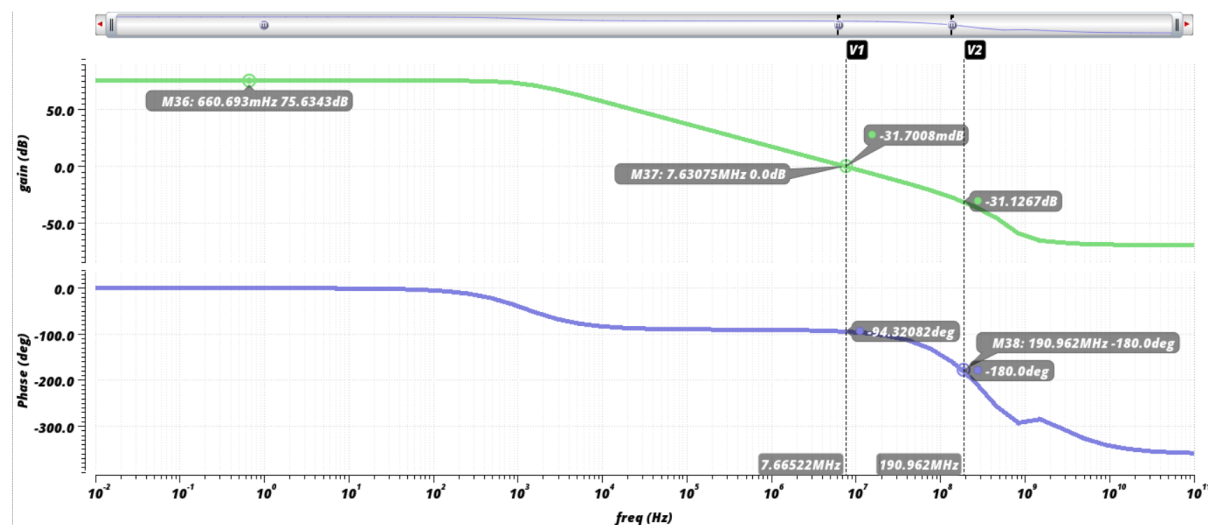


Figure 3: Differential Gain

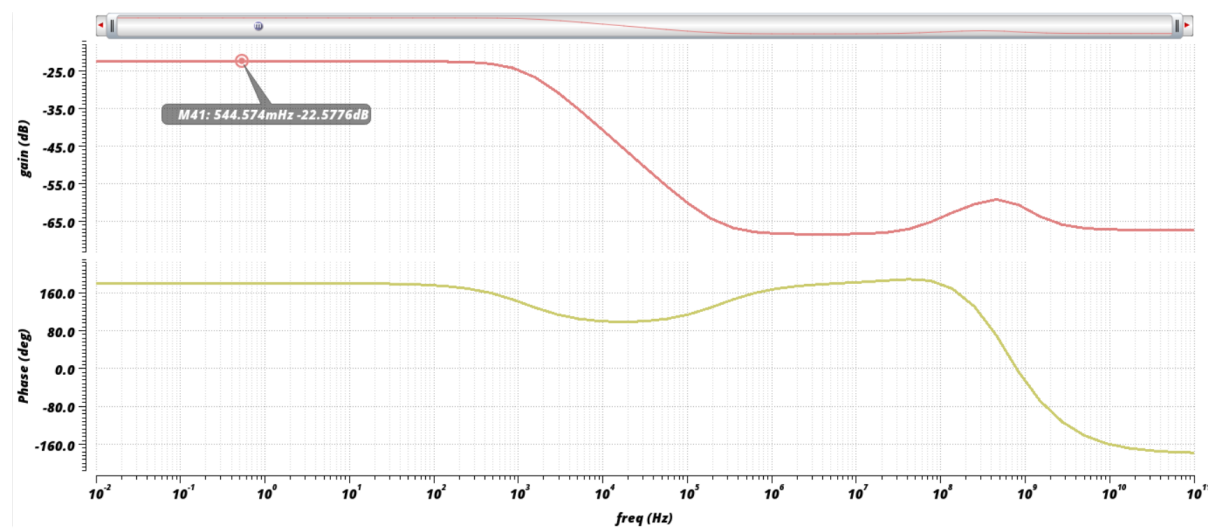


Figure 4: Common Mode Gain

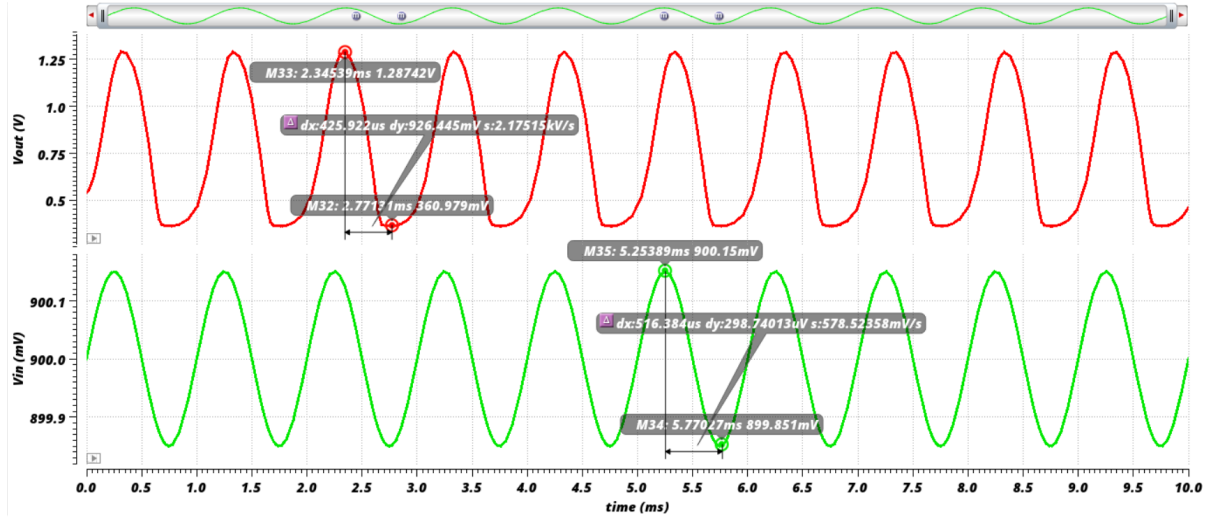


Figure 5: Transient (Swing)

Table for Different Process Corners

Process Corner	Gain (A_d)	CM Gain (A_{cm})	CMRR (dB)	Swing
TT	78.82	-28.83	106.7	926 mV
FS	72.4	-9.199	93.45	901 mV
SF	53.36	-27.01	90.37	823 mV
SS	81.2	-19.169	100.369	1.1 V
FF	57.4	-5.62	63.02	769 mV

Table 1

Conclusion

The folded cascode OTA with NMOS load exhibits robust performance in the TT and SS process corners, achieving high gain, CMRR, and output swing due to increased output resistance. Performance degrades in the SF and FF corners, where reduced gain and CMRR are observed because of mobility mismatch and lower output resistance. Overall, the design meets specifications mainly under nominal and slow-process conditions, indicating the need for careful biasing to ensure consistent performance across all corners.

TASK 2: Designing Folded Cascode OTA with PMOS Load

Design a Folded Cascode Operational Amplifier with a single-ended output for the following specifications. Use the Current reference to bias the transistors wherever required. Run the simulations for all corners (tt, ss, ff, sf, fs) and report the specs obtained for all corners in tabular format.

- Differential Gain > 80 dB
- CMRR > 100 dB
- ICMR: $0 - 1.2$ V
- Max. Output Swing: 1 V
- Power $< 20 \mu\text{W}$
- GBW > 10 MHz

Schematic of Folded CAscode with PMOS load

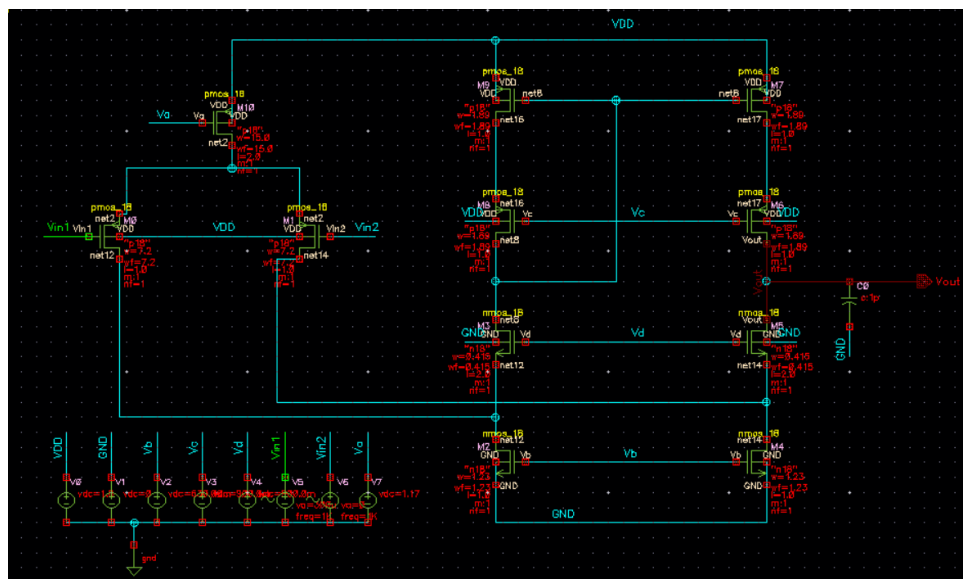


Figure 6: component parameters

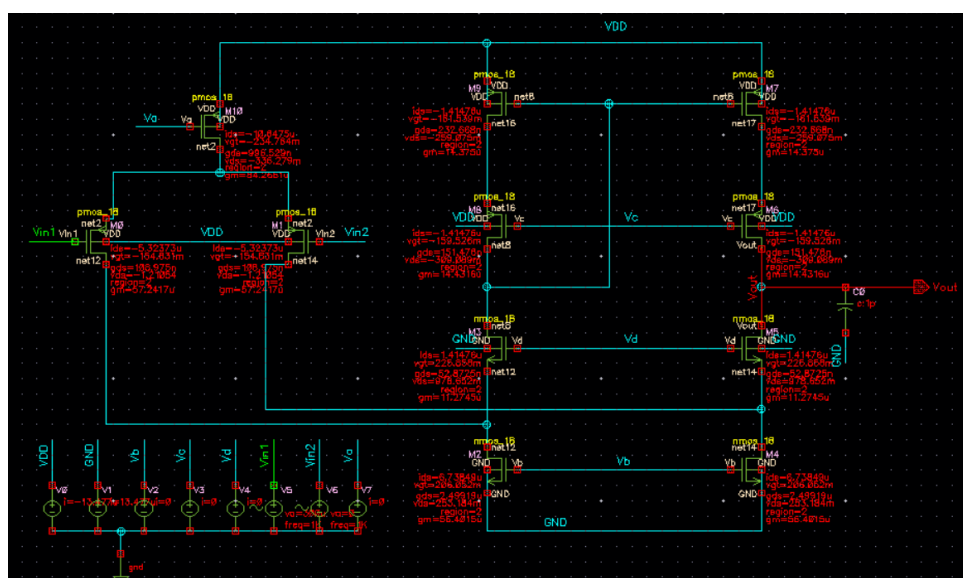


Figure 7: DC Parameters

Simulation Results

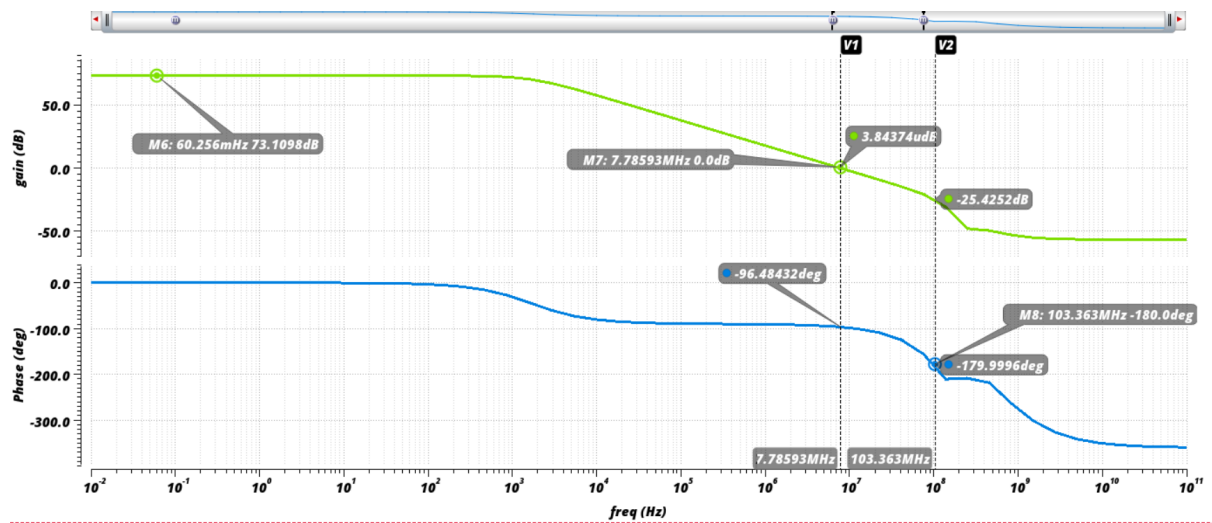


Figure 8: Differential Gain

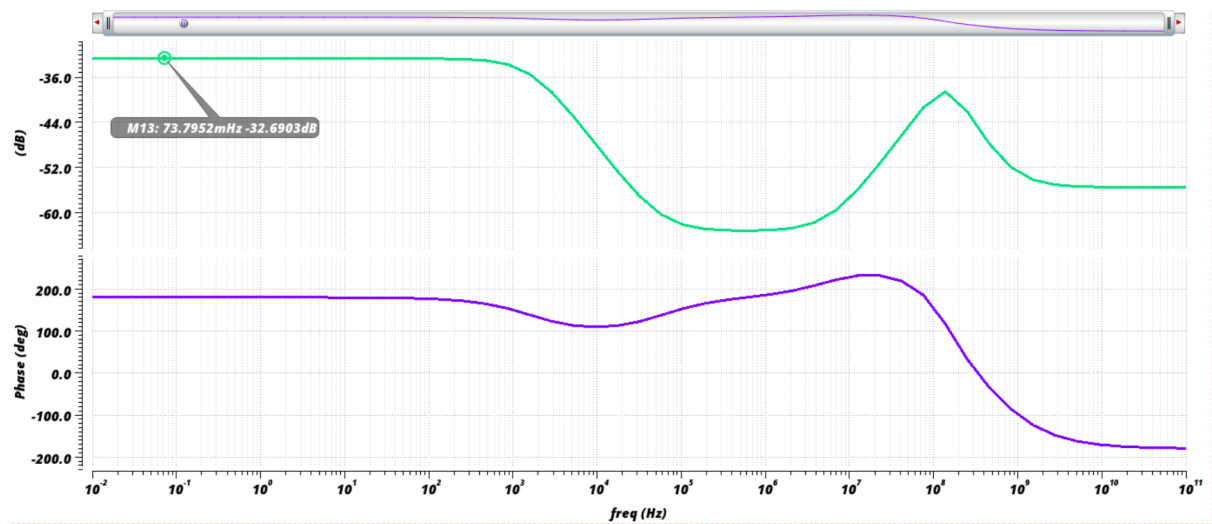


Figure 9: Common Mode Gain

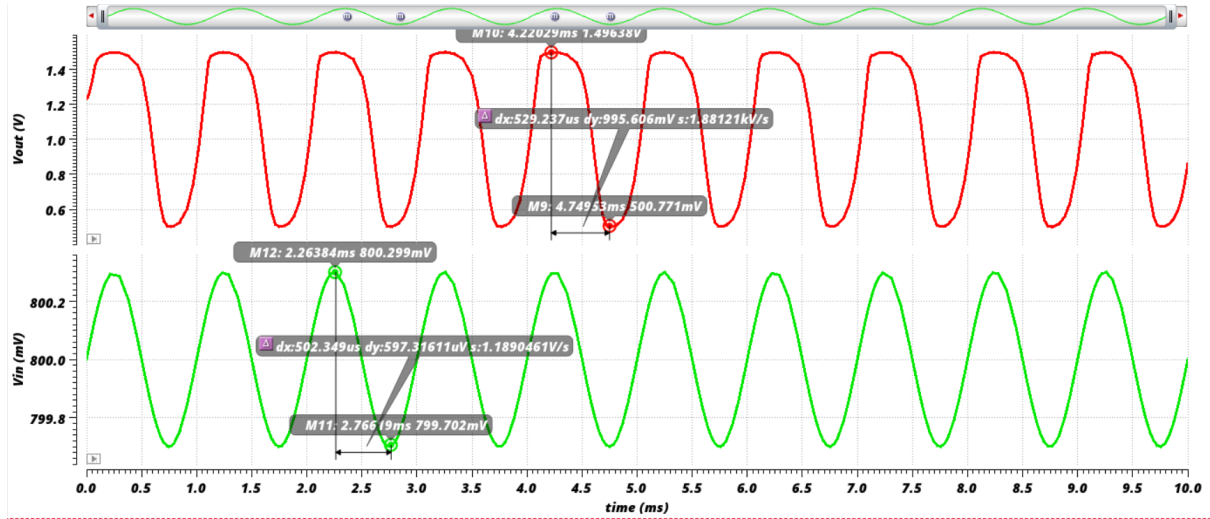


Figure 10: Transient (Swing)

Table for Different Process Corners

Process Corner	A_d	A_{cm}	CMRR (dB)	Swing
TT	73.19	-32.69	105.799	995 mV
FS	62.5	-47.05	109.55	401 mV
SF	69.0	-30.72	95.92	746 mV
SS	73.6	-43.3	117.3	973 mV
FF	61.05	-33.0	94.05	820 mV

Table 2

Conclusion

The folded cascode OTA with PMOS load exhibits strong performance across process corners, with the SS corner achieving the highest CMRR due to increased output resistance. The TT corner shows balanced gain, CMRR, and output swing, indicating stable nominal operation. Performance degradation is observed in the FS and FF corners, where reduced gain and output swing occur because of lower effective output resistance and device mismatch. Overall, the PMOS-loaded folded cascode provides better common-mode rejection across corners, while gain and swing remain sensitive to fast-process variations.